

A Rényi family in the entanglement-assisted asymptotic spectrum of graphs

Seth Douglas

Nidhal Mghirbi

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Abstract

For a finite confusability graph G and an order $\alpha \in [0, 2]$, we minimize the Petz–Rényi information radius over all classical–quantum channels whose output supports respect the nonedges of G . We prove that the resulting parameter ϑ_α is additive under disjoint union, multiplicative under strong product, normalized on the one-vertex graph, and monotone under entanglement-assisted cohomomorphisms. Thus every ϑ_α is an element of the entanglement-assisted asymptotic spectrum of graphs introduced by Li and Zuiddam. The proof of multiplicativity applies to arbitrary joint representations and is based on a nested-compression identity, not on an assumption that an optimizer factors.

The pentagon distinguishes infinitely many members of the family. More precisely, for $0 \leq \alpha \leq 1/2$ we obtain explicit two-sided bounds, and $\vartheta_\alpha(C_5) = 5/2$ for $1/2 \leq \alpha \leq 2$. A sparse sequence of orders gives infinitely many pairwise distinct spectral points, all distinct from the Lovász number. At the endpoints, $\vartheta_0 = \vartheta$, $\vartheta_1 = 2^{C_{\min}}$, and $\vartheta_2 = \Sigma$. Finally, an exact rational moment certificate gives a 32-vertex graph for which $C_{\min}(G) < \log_2 \Sigma(G)$, so the last two endpoints are distinct as graph parameters. The finite certificate is included with a deterministic verifier.

1 Introduction

The asymptotic spectrum of graphs packages graph parameters that are simultaneously additive under disjoint union, multiplicative under strong product, normalized, and monotone for an appropriate cohomomorphism preorder. Zuiddam used this framework for ordinary graph homomorphisms [Zui19]; Li and Zuiddam introduced its quantum and entanglement-assisted variants [LZ21]. In the strongest of these preorders, Li and Zuiddam proved that the Lovász number is a spectral point, reported no other examples, and asked whether it might even be the only point [LZ21, Theorem 18 and the discussion following Conjecture 19]. To our knowledge, the present construction is the first to produce points of the entanglement-assisted graph spectrum beyond the Lovász number, and hence the first infinite family in that spectrum.

Rényi radii have appeared previously in graph theory from several directions. Dalai used classical–quantum Rényi information radii to recover the Lovász bound as the zero-error endpoint of a sphere-packing construction [Dal13]; this does not give finite-order graph parameters in an asymptotic spectrum. Vrana showed that each already-known point of the ordinary graph spectrum admits a distribution-valued probabilistic refinement [Vra20]; that construction presupposes a scalar spectral point and concerns the ordinary cohomomorphism preorder. In the weaker quantum and commuting-quantum preorders, fractional-Haemers and tracial-Haemers type parameters provide further points [LZ21, GGL22]. Li and Zuiddam noted that the corresponding monotonicity was not known for their stronger entanglement-assisted preorder. The result here proves both multiplicativity

for arbitrary joint representations and monotonicity for that stronger preorder. We do not claim that this family exhausts the spectrum.

We construct a one-parameter family of points. Let $(\rho_x)_{x \in V(G)}$ be density operators such that $\rho_x \rho_y = 0$ whenever $x \neq y$ are nonadjacent in the confusability graph G . For the Petz–Rényi divergence D_α , define

$$\log_2 \vartheta_\alpha(G) = \inf_{\substack{(\rho_x), \sigma > 0 \\ \text{Tr } \sigma = 1}} \max_x D_\alpha(\rho_x \| \sigma), \quad 0 \leq \alpha \leq 2. \quad (1)$$

The output dimension is arbitrary and finite, and no attainment is assumed. Our first main result is the following.

Theorem 1.1 (Spectral-point theorem). *For every $\alpha \in [0, 2]$, the map $G \mapsto \vartheta_\alpha(G)$ belongs to Li and Zuiddam’s entanglement-assisted asymptotic spectrum $X(\mathcal{G}, \leq_*)$.*

The theorem combines three different mechanisms. Supported-state minimization converts (1) into a projection-packing formula involving the power compression

$$c_s(P; \sigma) = \text{Tr}_{\text{ran } P} (P \sigma^s P)^{1/s}, \quad s = 1 - \alpha, \quad (2)$$

with the logarithmic limit at $s = 0$. Nested compression gives an exact tower identity. Applied to the row spans of an arbitrary representation of a strong product, it proves the difficult direction of multiplicativity. A separate orthogonal-sum inequality proves disjoint-union additivity. Finally, a flagged-mixture argument and data processing prove monotonicity for entanglement-assisted cohomomorphisms. The last step is the reason for the range $\alpha \leq 2$.

The family is not a redundant parametrization. On the pentagon we prove

$$\sqrt{5} \left(\frac{\sqrt{5}}{2} \right)^{\alpha/(1-\alpha)} \leq \vartheta_\alpha(C_5) \leq \min \left\{ \frac{5}{2}, 5^{1/(2-2\alpha)} \right\} \quad (0 \leq \alpha \leq \tfrac{1}{2}), \quad (3)$$

and equality $\vartheta_\alpha(C_5) = 5/2$ on $[1/2, 2]$. Interpolation between trace-norm and Hilbert–Schmidt operator inequalities is the main ingredient. The bounds imply the following.

Theorem 1.2 (Infinitely many points). *For $k \geq 0$, put $\alpha_k = (8^{k+1} + 1)^{-1}$. The points $\vartheta, \vartheta_{\alpha_0}, \vartheta_{\alpha_1}, \dots$ are pairwise distinct elements of $X(\mathcal{G}, \leq_*)$.*

The endpoints have operational interpretations. We recover

$$\vartheta_0 = \vartheta, \quad \vartheta_1 = 2^{C_{\min}}, \quad \vartheta_2 = \Sigma, \quad (4)$$

where $C_{\min}$ is the smallest Holevo capacity of a compatible classical–quantum channel and Σ is the fractional common-dominator simulation cost of Duan and Winter [DW16]. The first two identities are information-radius formulas; the last uses the equivalence of supported Petz order two and supported max-relative entropy. Duan and Winter already proved multiplicativity of Σ for two fixed cq-graphs [DW16, Proposition 18]. The graph-level product statement needed here is stronger in quantifiers: a representation of $G \boxtimes H$ may be chosen jointly and need not factor through representations of the two graphs. The nested-compression argument handles precisely this issue.

To show that the last two endpoints are themselves different, we give a finite exact example.

Theorem 1.3 (A small endpoint separation). *There is an explicitly defined 32-vertex graph G_{32} such that*

$$2^{C_{\min}(G_{32})} \leq 6.186983 \dots < 6.194232 \dots \leq \Sigma(G_{32}). \quad (5)$$

Both outer bounds are certified by exact rational data included with this paper. In particular, $\vartheta_1 \neq \vartheta_2$ as graph parameters.

The lower bound on Σ comes from a dimension-independent moment relaxation. Its key observation is that the harmonic compression is a vector-state moment of a second commuting family of projections. The dual matrix is rational and its positive semidefiniteness is certified exactly by an integer factor plus a diagonally dominant integer residual. A numerical SDP solver was used to find the certificate but is not used to accept it.

We emphasize the limits of the paper. The new points all dominate the Lovász number, so they do not improve the Lovász upper bound on entanglement-assisted Shannon capacity and do not decide whether the Lovász number is the least spectral point. Orders above two, the exact pentagon curve on $(0, 1/2)$, and a classification of the spectrum remain open.

Organization. Section 2 fixes conventions. Supported minimization and product multiplicativity are proved in Sections 3 and 4. Disjoint-union additivity and entanglement-assisted monotonicity are proved in Sections 5 and 6. The pentagon is treated in Section 7. The endpoint separation and its certificate are described in Section 8. Additional certificate details are in Section 9.

2 Preliminaries

All graphs are finite and simple. They are interpreted as confusability graphs: distinct nonadjacent vertices must be represented by orthogonal states. The disjoint union is denoted $G \sqcup H$ and the strong product by $G \boxtimes H$. We set $\vartheta_\alpha(K_0) = 0$ for the empty graph.

2.1 Compatible states and Rényi radii

Definition 2.1. A family $(\rho_x)_{x \in V(G)}$ of density operators on a finite-dimensional Hilbert space is *compatible with G* if $\rho_x \rho_y = 0$ whenever $x \neq y$ are nonadjacent in G .

For a faithful density operator σ , the base-two Petz–Rényi divergence is

$$D_\alpha(\rho \parallel \sigma) = \frac{1}{\alpha - 1} \log_2 \text{Tr } \rho^\alpha \sigma^{1-\alpha} \quad (\alpha \neq 1), \quad (6)$$

with the continuous value $D(\rho \parallel \sigma)$ at $\alpha = 1$ and $D_0(\rho \parallel \sigma) = -\log_2 \text{Tr } \Pi_\rho \sigma$ at $\alpha = 0$. Definition (1) is unchanged if centers are allowed to be singular with the standard support convention: a singular center can be mixed with the maximally mixed state and the mixing parameter sent to zero.

The graph parameters $C_{\min}$ and Σ can be defined independently as

$$C_{\min}(G) = \inf_{(\rho_x) \text{ compatible}} \max_p \left[S \left(\sum_x p_x \rho_x \right) - \sum_x p_x S(\rho_x) \right], \quad (7)$$

$$\Sigma(G) = \inf \{ \text{Tr } T : T \geq \rho_x \ \forall x, \ (\rho_x) \text{ compatible} \}. \quad (8)$$

The infima range over arbitrary finite output dimensions.

2.2 The entanglement-assisted preorder

We use the complement convention of Li and Zuiddam [LZ21]. In confusability language, $G \leq_* H$ means that there are a nonzero positive semidefinite operator ω and operators $\omega_g^h \geq 0$ satisfying

$$\sum_{h \in V(H)} \omega_g^h = \omega \quad \text{for every } g, \quad (9)$$

$$\omega_g^h \omega_{g'}^{h'} = 0 \quad \text{if } g \neq g' \text{ are nonadjacent in } G, \text{ and } h = h' \text{ or } h \sim_H h'. \quad (10)$$

We normalize $\text{Tr } \omega = 1$ and work on $\text{supp } \omega$.

The entanglement-assisted asymptotic spectrum $X(\mathcal{G}, \leq_*)$ consists of maps ϕ from finite graphs to $[0, \infty)$ satisfying

$$\begin{aligned}\phi(G \sqcup H) &= \phi(G) + \phi(H), & \phi(G \boxtimes H) &= \phi(G)\phi(H), \\ \phi(K_1) &= 1, & G \leq_* H &\implies \phi(G) \leq \phi(H).\end{aligned}$$

3 Supported minimization and power compressions

Let $P \neq 0$ be a projection and let $\sigma > 0$. For $s \in [-1, 1] \setminus \{0\}$ define, on $\text{ran } P$,

$$K_s(\sigma, P) = (P\sigma^s P)^{1/s}, \quad Z_s(\sigma, P) = \text{Tr } K_s(\sigma, P), \quad (11)$$

and set

$$K_0(\sigma, P) = \exp(P(\ln \sigma)P), \quad Z_0(\sigma, P) = \text{Tr } K_0(\sigma, P). \quad (12)$$

The operators are extended by zero off $\text{ran } P$.

Proposition 3.1 (Supported-state variational formula). *For $0 \leq \alpha \leq 2$, $s = 1 - \alpha$, and $P \neq 0$,*

$$\min_{\substack{\rho \text{ state} \\ \text{supp } \rho \subseteq \text{ran } P}} D_\alpha(\rho \| \sigma) = -\log_2 Z_s(\sigma, P). \quad (13)$$

The minimum is attained at K_s/Z_s .

Proof. For $0 < \alpha < 1$, put $A = P\sigma^{1-\alpha}P$. Schatten Hölder gives

$$\text{Tr } \rho^\alpha A \leq (\text{Tr } A^{1/(1-\alpha)})^{1-\alpha}.$$

Because $(\alpha - 1)^{-1} < 0$, this is the required lower bound on the divergence, and equality holds for $\rho = A^{1/(1-\alpha)} / \text{Tr } A^{1/(1-\alpha)}$.

For $1 < \alpha \leq 2$, diagonalize $A = P\sigma^{1-\alpha}P$ on $\text{ran } P$. Scalar Jensen applied to the diagonal entries of ρ^α , followed by the scalar constrained minimum, gives

$$\text{Tr } A\rho^\alpha \geq (\text{Tr } A^{1/(1-\alpha)})^{1-\alpha},$$

with equality at the same state. This is also the reverse-Hölder calculation. At $\alpha = 1$, the Gibbs variational identity gives

$$D(\rho \| \sigma) = D\left(\rho \left\| \frac{K_0}{Z_0}\right.\right) - \log_2 Z_0.$$

At $\alpha = 0$, maximizing $\text{Tr } \Pi_\rho \sigma$ over states supported in P gives $\text{Tr } P\sigma = Z_1$. This proves all cases. $\square$

Consequently,

$$\frac{1}{\vartheta_\alpha(G)} = \sup_{(P_x), \sigma > 0} \min_x Z_{1-\alpha}(\sigma, P_x), \quad (14)$$

where (P_x) ranges over projection packings of G , with $P_x P_y = 0$ at distinct nonedges. Indeed, the supports of every compatible family form such a packing, while the minimizing states in [Theorem 3.1](#) form a compatible family for every packing.

Proposition 3.2 (Endpoint identities). *The parameters in (1) satisfy*

$$\vartheta_0 = \vartheta, \quad \vartheta_1 = 2^{C_{\min}}, \quad \vartheta_2 = \Sigma.$$

Moreover, $\alpha \mapsto \vartheta_\alpha(G)$ is nondecreasing.

Proof. At order zero, (14) is the standard projector form of the Lovász number [Lov79, Dal13]. At order one, the information-radius identity

$$\max_p \chi(p; \rho) = \inf_\sigma \max_x D(\rho_x \| \sigma)$$

and (7) give the second equality. At order two, $K_{-1} = (P\sigma^{-1}P)^{-1}$ is the shorted operator [AT75]. The same supported minimum is obtained with max-relative entropy. Normalizing a common dominator $T = t\sigma$ identifies the optimized max-radius with $\log_2 \Sigma(G)$, giving the third equality. Monotonicity follows from monotonicity of the Petz divergence in its order before taking infima. $\square$

The key algebraic fact is the following exact nesting property.

Lemma 3.3 (Tower identity). *If $P \leq Q$ are projections and $s \in [-1, 1]$, then*

$$K_s(K_s(\sigma, Q), P) = K_s(\sigma, P). \quad (15)$$

If $z = Z_s(\sigma, Q)$, then

$$Z_s\left(\frac{K_s(\sigma, Q)}{z}, P\right) = \frac{Z_s(\sigma, P)}{z}. \quad (16)$$

Proof. The outer occurrence of K_s is evaluated on the ambient space $\text{ran } Q$; on that space $K_s(\sigma, Q)$ is faithful. For $s \neq 0$, the s th power of $K_s(\sigma, Q)$ inside $\text{ran } Q$ is $Q\sigma^s Q$. Compression to $P \leq Q$ gives $P\sigma^s P$, proving (15). For $s = 0$, take logarithms inside $\text{ran } Q$. Equation (16) follows from degree-one homogeneity in the first argument. $\square$

4 Strong-product multiplicativity

The easy product inequality follows by tensoring near-optimal witnesses. The reverse inequality must allow a representation of $G \boxtimes H$ that is not a tensor product.

Theorem 4.1 (Arbitrary-joint product theorem). *For $0 \leq \alpha \leq 2$ and finite graphs G, H ,*

$$\vartheta_\alpha(G \boxtimes H) = \vartheta_\alpha(G) \vartheta_\alpha(H). \quad (17)$$

Proof. If either graph is empty, both sides vanish by the convention $\vartheta_\alpha(K_0) = 0$. Assume henceforth that both are nonempty. Fix $s = 1 - \alpha$ and take a joint packing (P_{xy}) of $G \boxtimes H$ with a faithful center σ . Suppose $Z_s(\sigma, P_{xy}) \geq 2^{-r}$ for every pair (x, y) . For each x , let S_x be the projection onto the span of the ranges of $(P_{xy})_y$ and put $z_x = Z_s(\sigma, S_x)$.

If x, x' are distinct nonadjacent vertices of G , then every $(x, y), (x', y')$ is a nonedge of the strong product. Hence $S_x S_{x'} = 0$, so (S_x) is a packing of G . For each fixed x , $(P_{xy})_y$ is a packing of H on $\text{ran } S_x$. By (16), its reduced center $K_s(\sigma, S_x)/z_x$ satisfies

$$Z_s\left(\frac{K_s(\sigma, S_x)}{z_x}, P_{xy}\right) = \frac{Z_s(\sigma, P_{xy})}{z_x} \geq \frac{2^{-r}}{z_x}.$$

The packing formula for H therefore implies $\vartheta_\alpha(H) \leq 2^r z_x$, or $z_x \geq 2^{\log_2 \vartheta_\alpha(H) - r}$. Applying the packing formula once more to (S_x) gives

$$\log_2 \vartheta_\alpha(G) \leq r - \log_2 \vartheta_\alpha(H).$$

Thus every joint radius r is at least the sum of the two graph radii. Taking the infimum proves the reverse inequality.

For the forward inequality, tensor packings and centers. The traces in (11) multiply under tensor products, so near-optimal witnesses give the product upper bound. No attainment is required. $\square$

At $\alpha = 2$, Theorem 4.1 gives $\Sigma(G \boxtimes H) = \Sigma(G)\Sigma(H)$. This removes the regularization from the graph-level no-signalling simulation quantity considered by Duan and Winter [DW16]; their fixed-channel multiplicativity does not by itself imply the graph-level statement because the optimizing joint representation need not factor.

5 Disjoint-union additivity

We first record the compression identity in divergence form.

Lemma 5.1 (Compression identity). *Let $Q \neq 0$ be a projection, $s = 1 - \alpha \in [-1, 1]$, and $\sigma_Q = K_s(\sigma, Q)/Z_s(\sigma, Q)$. Every state ρ supported in Q satisfies*

$$D_\alpha(\rho \parallel \sigma) = D_\alpha(\rho \parallel \sigma_Q) - \log_2 Z_s(\sigma, Q). \quad (18)$$

Proof. For $\alpha \notin \{0, 1\}$, insert $Q\sigma^s Q = Z_s^s \sigma_Q^s$ into the Petz moment. At $\alpha = 0$ use $Q\sigma Q = Z_1 \sigma_Q$ under the support projection of ρ . At $\alpha = 1$, use $Q(\ln \sigma)Q = \ln \sigma_Q + (\ln Z_0)Q$. $\square$

Lemma 5.2 (Orthogonal-sum inequality). *If $Q_1, \dots, Q_m$ are mutually orthogonal projections, $\sigma > 0$, and $s \leq 1$, then*

$$\sum_j Z_s(\sigma, Q_j) \leq 1. \quad (19)$$

Proof. For $s \neq 0$, set $A = \sigma^s$ and $f(t) = t^{1/s}$; for $s = 0$, set $A = \ln \sigma$ and $f(t) = e^t$. In every case used here, f is nonnegative and convex on the spectrum of A , and $\text{Tr } f(A) = 1$. Diagonalize each $Q_j A Q_j$ in an orthonormal basis $(e_{jk})_k$ of $\text{ran } Q_j$. Scalar Jensen for the spectral measure of A in e_{jk} gives

$$f(\langle e_{jk}, A e_{jk} \rangle) \leq \langle e_{jk}, f(A) e_{jk} \rangle.$$

Summing over the orthonormal family $(e_{jk})_{j,k}$ proves (19). $\square$

Theorem 5.3 (Disjoint-union additivity). *For $0 \leq \alpha \leq 2$,*

$$\vartheta_\alpha(G \sqcup H) = \vartheta_\alpha(G) + \vartheta_\alpha(H).$$

Proof. The assertion is immediate if either graph is empty, so assume both are nonempty. For the upper bound, take near-optimal families on orthogonal Hilbert-space summands with centers σ_G, σ_H and radius bounds $\log_2 a$ and $\log_2 b$. The block center $(a\sigma_G \oplus b\sigma_H)/(a+b)$ has radius at most $\log_2(a+b)$ by Theorem 5.1.

Conversely, take a compatible family of $G \sqcup H$ with center σ and radius at most λ . Let Q_G, Q_H be the support projections of the two subfamilies. Cross pairs are nonedges, so $Q_G Q_H = 0$. By Theorem 5.1,

$$\vartheta_\alpha(G) \leq 2^\lambda Z_s(\sigma, Q_G), \quad \vartheta_\alpha(H) \leq 2^\lambda Z_s(\sigma, Q_H).$$

The sum is at most 2^λ by Theorem 5.2. Infimize over λ . $\square$

In particular, $\vartheta_\alpha(\overline{K_N}) = N$ and $\vartheta_\alpha(K_1) = 1$.

6 Entanglement-assisted monotonicity

Lemma 6.1 (Flagged mixtures). *Let $\omega_h \geq 0$ with $\sum_h \omega_h = \omega$ and $\text{Tr } \omega = 1$. Let τ_h be states and let $\sigma > 0$ be a density operator. If $D_\alpha(\tau_h \| \sigma) \leq \lambda$ whenever $\omega_h \neq 0$, then, for $0 \leq \alpha \leq 2$,*

$$D_\alpha \left(\sum_h \omega_h \otimes \tau_h \middle\| \omega \otimes \sigma \right) \leq \lambda. \quad (20)$$

Proof. Write $w_h = \text{Tr } \omega_h$ and $\hat{\omega}_h = \omega_h / w_h$, omitting zero terms. Add a classical flag and consider

$$\begin{aligned} \Omega &= \sum_h w_h |h\rangle\langle h| \otimes \hat{\omega}_h \otimes \tau_h, \\ \Omega' &= \sum_h w_h |h\rangle\langle h| \otimes \hat{\omega}_h \otimes \sigma. \end{aligned}$$

The Petz moment is block additive, and the common first two tensor factors cancel. For $0 < \alpha < 1$ each block moment is at least $2^{(\alpha-1)\lambda}$; for $1 < \alpha \leq 2$ the inequalities reverse. The order-one expression is the weighted average of the relative entropies, and the order-zero expression is handled by support projections. Thus $D_\alpha(\Omega \| \Omega') \leq \lambda$.

Discarding the flag gives (20). Petz data processing holds for $\alpha \in (0, 1) \cup (1, 2]$ [Pet86, Tom16]; order one follows by continuity. At order zero, $\Pi_\Omega \leq I \otimes \Pi_{\text{Tr}_F \Omega}$ implies directly that the order-zero divergence cannot increase under the partial trace. $\square$

Theorem 6.2 (Entanglement-assisted monotonicity). *If $G \leq_* H$, then $\vartheta_\alpha(G) \leq \vartheta_\alpha(H)$ for every $\alpha \in [0, 2]$.*

Proof. If G is empty the claim is immediate. If G is nonempty, the defining data force H to be nonempty, so the construction below applies. Take the data ω, ω_g^h from (9)–(10), and a compatible target family (τ_h) with faithful center σ . Define

$$\tau'_g = \sum_h \omega_g^h \otimes \tau_h, \quad \sigma' = \omega \otimes \sigma$$

on $\text{supp } \omega \otimes \mathcal{H}$. The states are normalized. If g, g' are distinct nonadjacent vertices, then every summand of $\tau'_g \tau'_{g'}$ vanishes: condition (10) kills the terms with $h = h'$ or $h \sim_H h'$, while target compatibility kills the remaining terms. Thus (τ'_g) is compatible with G . Applying Theorem 6.1 for each g shows that its radius is at most the target radius. Take infima. $\square$

Proof of Theorem 1.1. For K_1 , the packing formula is attained with $P = I$, giving $Z_s(\sigma, I) = 1$, while the one-projection case of Theorem 5.2 gives $Z_s(\sigma, P) \leq 1$ for every witness. Thus $\vartheta_\alpha(K_1) = 1$. Additivity is Theorem 5.3, multiplicativity is Theorem 4.1, and monotonicity is Theorem 6.2. $\square$

Petz data processing generally fails above order two. Accordingly, the present argument makes no spectral claim for $\alpha > 2$.

7 The pentagon and infinitely many points

Index C_5 by $\mathbb{Z}_5$ with $i \sim i \pm 1$. In our confusability convention, a pentagon packing satisfies $P_i P_{i+2} = 0$.

Lemma 7.1 (Two endpoint operator inequalities). *For every pentagon packing and every operator X ,*

$$\sum_i \|P_i X P_i\|_1 \leq \sqrt{5} \|X\|_1, \quad (21)$$

$$\sum_i \|P_i X P_i\|_2^2 \leq 2 \|X\|_2^2. \quad (22)$$

Proof. For a rank-one operator $X = ab^*$, $\|P_i ab^* P_i\|_1 = \|P_i a\| \|P_i b\|$. The projector form of $\vartheta(C_5) = \sqrt{5}$ gives $\sum_i \|P_i v\|^2 \leq \sqrt{5} \|v\|^2$; Cauchy–Schwarz therefore proves (21) for rank one. A singular-value decomposition and the triangle inequality prove it for general X .

For (22), put $T = \sum_i P_i$. In the convention $P_i P_{i+2} = 0$, the two surviving projections P_{i-1} and P_{i+1} are orthogonal. Hence

$$P_i T P_i = P_i + P_i(P_{i-1} + P_{i+1})P_i \leq 2P_i. \quad (23)$$

Let

$$\mathcal{A} : \bigoplus_i P_i \mathcal{B}_2 P_i \longrightarrow \mathcal{B}_2, \quad \mathcal{A}((X_i)_i) = \sum_i X_i.$$

For $X_i = P_i X_i P_i$, Cauchy–Schwarz and $ab \leq (a^2 + b^2)/2$ give

$$|\operatorname{Tr} X_i^* X_j| \leq \frac{1}{2} (\|P_j X_i\|_2^2 + \|X_j P_i\|_2^2).$$

After summing over i, j and using (23),

$$\left\| \sum_i X_i \right\|_2^2 \leq \frac{1}{2} \sum_i [\operatorname{Tr}(X_i^* T X_i) + \operatorname{Tr}(X_i T X_i^*)] \leq 2 \sum_i \|X_i\|_2^2.$$

Thus $\|\mathcal{A}\|^2 \leq 2$. Since $\mathcal{A}^*(X) = (P_i X P_i)_i$, the same bound for $\mathcal{A}^*$ is exactly (22). $\square$

Lemma 7.2 (Interpolated pentagon inequality). *Let $s \in [1/2, 1]$ and $p = 1/s$. For every faithful state σ and every pentagon packing,*

$$\sum_i Z_s(\sigma, P_i) \leq B(p) := 5^{(2-p)/2} 2^{p-1}. \quad (24)$$

Proof. The map $\mathcal{T} : X \mapsto (P_i X P_i)_i$ has norm at most $\sqrt{5}$ from S_1 to $\ell_1(S_1)$ by (21), and norm at most $\sqrt{2}$ from S_2 to $\ell_2(S_2)$ by (22). Complex interpolation for Schatten direct sums [BL76, Bha97] gives, for $1 \leq p \leq 2$,

$$\left(\sum_i \|P_i X P_i\|_p^p \right)^{1/p} \leq 5^{(2-p)/(2p)} 2^{(p-1)/p} \|X\|_p.$$

One may obtain this directly from the three-lines function built from the polar decompositions of X and the dual block operator. Put $X = \sigma^s$. Then $\|X\|_p^p = \operatorname{Tr} \sigma = 1$ and $\|P_i \sigma^s P_i\|_p^p = Z_s(\sigma, P_i)$. Raising the interpolated bound to the p th power proves (24). $\square$

Theorem 7.3 (Pentagon bounds). *Put $r = \alpha/(1 - \alpha)$. For $0 \leq \alpha \leq 1/2$,*

$$\sqrt{5} \left(\frac{\sqrt{5}}{2} \right)^r \leq \vartheta_\alpha(C_5) \leq \min \left\{ \frac{5}{2}, \sqrt{5} 5^{r/2} \right\}. \quad (25)$$

For $1/2 \leq \alpha \leq 2$, $\vartheta_\alpha(C_5) = 5/2$.

Proof. The pentagon is vertex-transitive. Given any witness, take the orthogonal direct sum of its relabelings by all graph automorphisms and weight the center equally across the summands. Degree-one homogeneity makes every vertex value in the symmetrized witness equal to the average of the original five values; the reverse inequality follows from minimum at most average. Hence $1/\vartheta_\alpha(C_5) = \sup_{P,\sigma} \frac{1}{5} \sum_i Z_{1-\alpha}(\sigma, P_i)$. With $p = (1-\alpha)^{-1}$, [Theorem 7.2](#) gives the lower bound $5/B(p)$, which is the left side of [\(25\)](#).

For the uniform upper bound $5/2$, use on $\mathbb{C}^5$ the commuting states

$$\rho_i = \frac{1}{2}(|i\rangle\langle i| + |i+1\rangle\langle i+1|), \quad \sigma = I/5.$$

Their supports satisfy the pentagon nonedge relations, and every Petz order has radius $\log_2(5/2)$.

For the other upper bound, let

$$u_i = \left(\cos \gamma, \sin \gamma \cos \frac{2\pi i}{5}, \sin \gamma \sin \frac{2\pi i}{5} \right), \quad \cos^2 \gamma = 1/\sqrt{5}.$$

Then $u_i \perp u_{i+2}$. Against a faithful approximation to $|c\rangle\langle c|$, $c = (1, 0, 0)$, the pure-state Petz radius tends to $\log_2 \sqrt{5}/(1-\alpha)$, giving $\vartheta_\alpha(C_5) \leq 5^{1/(2-2\alpha)}$.

At $\alpha = 1/2$, the lower bound is $5/2$. Monotonicity in α and the uniform upper bound give equality throughout $[1/2, 2]$. $\square$

In particular, $\vartheta_\alpha(C_5) > \sqrt{5}$ for every $\alpha > 0$, while the pure-handle upper bound is below $5/2$ for

$$\alpha < 1 - \frac{\log 5}{2 \log(5/2)} = 0.121765\dots$$

Proof of [Theorem 1.2](#). Let $r_k = 8^{-(k+1)}$ and $\alpha_k = r_k/(1+r_k)$. By [Theorem 7.3](#),

$$\begin{aligned} \vartheta_{\alpha_{k+1}}(C_5) &\leq \sqrt{5} 5^{r_k/16}, \\ \vartheta_{\alpha_k}(C_5) &\geq \sqrt{5} (\sqrt{5}/2)^{r_k}. \end{aligned}$$

The integer inequality $2^{16} = 65536 < 78125 = 5^7$ is equivalent to $5^{1/16} < \sqrt{5}/2$, so each displayed upper bound is strictly below the next displayed lower bound. Also $\vartheta_{\alpha_0}(C_5) \leq 5^{9/16} < 5/2$, and every value is strictly above $\sqrt{5}$. Hence all listed spectral points are distinct. $\square$

Combining [Theorem 1.2](#) with [Theorem 1.3](#) shows that ϑ_0 , the countable ladder $(\vartheta_{\alpha_k})_k$, ϑ_1 , and ϑ_2 are pairwise distinct points of $X(\mathcal{G}, \leq_*)$.

8 A 32-vertex endpoint separation

We now define the graph and state the exact finite certificate. This section is logically independent of [Theorem 4.1](#): it proves the finite strict gap directly from the operational definitions [\(7\)](#) and [\(8\)](#).

8.1 The magic-square seed and its clone graph

Index the nine cells of a 3×3 array by $(i, j) \in \{0, 1, 2\}^2$. There are six contexts: the three rows and three columns. A vertex is a context together with a sign assignment to its three cells. Row assignments have product $+1$; column assignments have products $(+1, +1, -1)$. Each context therefore has four vertices and the seed F has 24 vertices. Two distinct vertices are declared orthogonal if they belong to the same context or assign opposite signs to a shared cell. This gives 216 ordered forbidden pairs.

The standard two-qubit packing uses the Mermin–Peres array

$$\begin{pmatrix} Z \otimes I & I \otimes Z & Z \otimes Z \\ I \otimes X & X \otimes I & X \otimes X \\ Z \otimes X & X \otimes Z & Y \otimes Y \end{pmatrix}. \quad (26)$$

For a context whose first two commuting observables are A, B , the vertex with first two signs (s, t) is represented by

$$Q_{A,B;s,t} = \frac{1}{4}(I + sA)(I + tB). \quad (27)$$

These are rank-one projections. Fix the first-row $(+, +)$ vertex 0, so $Q_0 = |00\rangle\langle 00|$, and define $j_x = 4 \operatorname{Tr} Q_x Q_0$. The four classes $j = 0, 1, 2, 4$ have sizes 9, 8, 6, 1.

For a profile $k = (k_0, k_1, k_2, k_4)$, replace every class- j vertex by k_j mutually nonadjacent clones and delete it if $k_j = 0$. Each clone inherits all adjacencies of its parent to vertices outside its clone class. The graph

$$G_{32} = F[(2, 1, 1, 0)] \quad (28)$$

has $2 \cdot 9 + 8 + 6 = 32$ vertices.

8.2 Gibbs and harmonic support weights

For a faithful state σ and a projection P , put

$$w_\sigma(P) = \operatorname{Tr} \exp(P(\ln \sigma)P), \quad (29)$$

$$h_\sigma(P) = \operatorname{Tr} \sigma_{/P}, \quad (30)$$

where $\sigma_{/P}$ is the largest positive operator below σ whose range is contained in $\operatorname{ran} P$. These are exactly $Z_0(\sigma, P)$ and $Z_{-1}(\sigma, P)$.

The certificate has two halves. A rational direct sum of the packing (26) and one-dimensional clique packings produces a Gibbs vector and therefore a compatible cq channel with small information radius. A moment-relaxation dual bounds every possible harmonic vector, in every dimension, and therefore every common-dominator simulation.

Proposition 8.1 (Direct operational bridge). *Let $F[k]$ be a clone graph.*

1. *If a packing and state have Gibbs weights w_x , then*

$$2^{C_{\min}(F[k])} \leq \left(\min_{x:k_x > 0} \frac{w_x}{k_x} \right)^{-1}.$$

2. *Suppose every packing and state satisfy a linear harmonic inequality $\sum_x \mu_x h_\sigma(P_x) \leq \lambda$, with $\mu_x \geq 0$. Then*

$$\Sigma(F[k]) \geq \frac{\sum_x \mu_x k_x}{\lambda}.$$

Proof. For the first part, let $A = \prod_{x:k_x > 0} [k_x]$ and work on $\bigoplus_{f \in A} \mathcal{H}$. Put

$$\sigma' = |A|^{-1} \bigoplus_{f \in A} \sigma, \quad P'_{x,a} = \bigoplus_{f \in A: f(x)=a} P_x,$$

with zero blocks in the omitted summands. Distinct clones of one parent are orthogonal, and external forbidden pairs remain orthogonal because clones inherit their parent's relations. Functional calculus

on the direct sum gives $w_{\sigma'}(P'_{x,a}) = w_x/k_x$. On a support P , the Gibbs state $\rho = \exp(P \ln \sigma P)/w_\sigma(P)$ satisfies $D(\rho\|\sigma) = -\log_2 w_\sigma(P)$. The information-radius identity therefore gives the bound.

For the second part, take an arbitrary feasible common-dominator tuple $T \geq \rho_{x,a}$ and set $t = \text{Tr } T$, $\sigma = T/t$. Each $\rho_{x,a}/t$ is feasible in the shorted-operator maximization, so $h_\sigma(P_{x,a}) \geq 1/t$. The clone supports are orthogonal. If $P_x = \sum_a P_{x,a}$, the trace of the shorted operator is superadditive over these blocks; this follows from the Schur-complement formula for the diagonal blocks of $(P_x \sigma^{-1} P_x)^{-1}$. Hence $h_\sigma(P_x) \geq k_x/t$. The assumed inequality gives $t \geq (\sum_x \mu_x k_x)/\lambda$. Singular T follows by faithful regularization and a limit. $\square$

8.3 The dimension-independent moment relaxation

Let (P_x) be any packing and $\sigma > 0$. Set $\Omega = \text{vec}(\sqrt{\sigma})$. Let F_x be the orthogonal projection onto $\sigma^{-1/2} \text{ran } P_x$ and set $E_x = F_x^\top$.

Lemma 8.2 (Harmonic weight as a moment). *For every x ,*

$$(P_x \otimes E_x)\Omega = (I \otimes E_x)\Omega, \quad \langle \Omega, I \otimes E_x \Omega \rangle = h_\sigma(P_x). \quad (31)$$

Proof. The range of $\sqrt{\sigma}F_x$ is $\text{ran } P_x$, proving the vector identity. Moreover $\sqrt{\sigma}F_x\sqrt{\sigma}$ is below σ and supported in P_x . If $0 \leq A \leq \sigma$ has range in P_x , then $\sigma^{-1/2}A\sigma^{-1/2}$ is a contraction supported in F_x , hence is at most F_x . Thus $\sqrt{\sigma}F_x\sqrt{\sigma} = \sigma_{/P_x}$. Taking the vector-state expectation proves the second identity. $\square$

The two projection families $P_x \otimes I$ and $I \otimes E_y$ commute, but no orthogonality relation is imposed on the E_y . Form a Gram matrix from the 409 monomials

$$1, \quad (P_a \otimes I)_{a=0}^{23}, \quad (I \otimes E_b)_{b=0}^{23}, \quad (P_a \otimes E_b)_{a=b \text{ or } a \not\leq b}.$$

Its entries are moments of words in the two families. Idempotence, forbidden adjacent P pairs, and the two endpoint consequences of (31) give sound moment identifications. Averaging over the 48 seed automorphisms fixing vertex zero and taking real parts preserves positive semidefiniteness. The construction is a finite, symmetry-reduced noncommutative moment relaxation in the spirit of [NPA08, Mgh26]. The resulting table has 1009 moment keys.

Let $\Gamma(y)$ be the resulting real symmetric matrix, with one variable per key. The primal relaxation maximizes t subject to

$$\Gamma(y) \succeq 0, \quad y_1 = 1, \quad y_x \geq tk_x \quad (k_x > 0). \quad (32)$$

By Theorem 8.2, every finite-dimensional packing and state gives a feasible point. Therefore the relaxation upper bounds the true harmonic radius.

A dual certificate consists of $Z \succeq 0$, multipliers $\mu_x \geq 0$, and λ such that, for each moment key κ ,

$$\sum_{(i,j):\text{key}(i,j)=\kappa} Z_{ij} + \sum_{x:\kappa_x=\kappa} \mu_x = \lambda [\kappa = 1]. \quad (33)$$

Weak duality then gives

$$t_H(F[k]) \leq \frac{\lambda}{\sum_x \mu_x k_x}, \quad \Sigma(F[k]) \geq \frac{\sum_x \mu_x k_x}{\lambda}. \quad (34)$$

8.4 Certified values

The included exact data give the following bounds.

graph	profile (k_0, k_1, k_2, k_4)	vertices	$2^{C_{\min}} \leq$	$\Sigma \geq$
G_{32}	$(2, 1, 1, 0)$	32	6.186983...	6.194232...
G_{44}	$(1, 2, 3, 1)$	44	8.949075...	8.962394...
G_{72}	$(4, 3, 2, 0)$	72	105/8	13.191800...

For G_{32} the exact radial quantities are

$$t_W \geq \frac{5225365592240894}{32329248967670751} > \frac{15147154828811}{93824992214030} \geq t_H. \quad (35)$$

The strict rational comparison and [Theorem 8.1](#) prove [Theorem 1.3](#). The analogous harmonic upper radii are

$$\frac{31406288103814}{281475531618259} \quad \text{and} \quad \frac{21337116511217}{281474976708840}$$

for G_{44} and G_{72} , respectively. The certified bit gaps are 0.001689, 0.002146, and 0.007324 bits.

9 Certificate verification

The file `certificates/small_graph_certificates.json` contains, for each graph, the rational Gibbs mixture, the rational multipliers μ , and the nonzero orbits of the dual matrix Z . The verifier performs the following steps.

1. It rebuilds the 24-vertex seed, all 216 ordered forbidden pairs, the two-qubit packing, the four overlap classes, and the 48 automorphisms.
2. It constructs the 409 monomials and the 1009-key moment table from the sound rewrite rules used above.
3. It expands the orbit data, checks every affine equation [\(33\)](#) in rational arithmetic, checks $\mu \geq 0$, and recomputes the Gibbs witness exactly.
4. To prove $Z \succeq 0$, it clears denominators and proposes an integer matrix L by rounding a floating-point Cholesky factor at scale 2^{40} . It then checks in exact integers that

$$R = 2^{80} DZ - DLL^T$$

is symmetric diagonally dominant with nonnegative diagonal. Hence $R \succeq 0$ and $Z = LL^T/2^{80} + R/(2^{80}D) \succeq 0$. Floating-point Cholesky success is not an acceptance condition.

The verifier has two independent negative controls. One damages an affine moment equation. The other changes only entries attached to identically zero moments, preserving every affine equation while breaking the exact positivity test. Both must be rejected.

10 Discussion and open questions

Theorem 1.1 supplies infinitely many points of the entanglement-assisted asymptotic spectrum beyond the Lovász number. It does not alter the spectral upper bound on entanglement-assisted Shannon capacity because every point in the present family is at least ϑ . Thus the possibility that ϑ is the least spectral point remains open.

Several concrete questions remain.

1. Does entanglement-assisted monotonicity extend beyond Petz order two? The flagged-mixture proof cannot: it fails already on finite examples at order three.
2. Is $\alpha \mapsto \vartheta_\alpha(C_5)$ continuous, and what is its exact value on $(0, 1/2)$? Continuity would immediately turn the countable family proved here into an uncountable one.
3. Does the limit as $\alpha \rightarrow \infty$ equal the complement of projective rank? Flat representations give the corresponding upper bound, but a dimension-independent near-packing argument is missing.
4. How small can a graph with $C_{\min} < \log_2 \Sigma$ be? The certificate proves existence on 32 vertices, not minimality.
5. Can the rational moment certificate be replaced by a short analytic inequality, and is the first relaxation level exact for the three examples?

Reproducibility. The certificate data and deterministic verification scripts accompany the paper. Their role is limited to the finite rational assertions in Sections 8 and 9. The all-dimensional arguments in Sections 3 to 6 are mathematical proofs, not consequences of numerical tests.

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